MILITARY SPECIFICATION

PROJECTILE, 60MM, SMOKE, WP, M302
METAL PARTS ASSEMBLY

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements and quality assurance provisions for the fabrication of parts, assemblies and packing of projectile assembly, 60MM, Smoke, WP, M302.

2. APPLICABLE DOCUMENTS

2.1. Government documents.

2.1.1 Specifications and standards. The following specifications and standards, form a part of this specification to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-490	Cleaning Methods and Pre-treatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT DRAWINGS (See 6.5)

PRODUCT AND PACKAGING DRAWINGS

10534894	- Projectile, 60MM, SMOKE, WP, M302 Metal Parts Assembly
7548319	- Box, Fiberboard Ammunition for Interplant Shipment of Projectiles, Empty

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B 46.1 - Surface Texture

(Application for copies should be addressed to: American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E-8 - Tension Testing of Metallic Materials

(Copies may be purchased directly from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials, shall be in accordance with the applicable specifications drawings and this specification.

3.2 Parts and assemblies. Parts and the projectile assembly shall comply with all requirements specified on drawing (dwg.) 10534894 and associated drawings, and with all requirements specified in applicable specifications and standards. Threads and other surfaces shall be free of burrs or foreign matter which may interfere with further assembly or proper functioning.

3.3 Forging. Each projectile body in a lot shall be heated above the upper critical temperature, oil quenched and tempered uniformly to meet the mechanical properties specified on the applicable drawing. No work involving plastic flow is permitted after tempering unless followed by a stress relief.

3.4 Cold shaping. Cold worked bodies shall be stress relieved at a minimum temperature of 700 degrees F. No work involving plastic flow of metal is permitted after final stress relief.

3.5 Mechanical properties. All components shall meet the mechanical properties requirements as specified on the applicable drawings.

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3.5.1 Adapter. Minimum mechanical properties as required by the applicable drawing may apply to specimens machined from the raw material, provided that the mechanical properties are not altered during manufacture.

3.6 Metal defects.

3.6.1 Body. The body, including the cavity, shall be free from cracks, splits, bursts, solid shuts, pipe, porosity, inclusions, folds, seams and other similar defects.

3.6.1.1 Body cavity. The body cavity shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits, and sharp edges.

3.6.2 Fuze adapter. The fuze adapter shall be free from cracks, splits, cold shuts, laminations, or other similar defects.

3.6.3 Visual standards (internal and external). Visual standards for evaluation of metal defects shall be established for each contract after the first month's production or whenever requested by the procuring activity. However, the establishment of visual standards shall not waive the requirement for compliance with 3.6.1 and 3.6.2.

3.7 Brazed joint. Braze metal must be visible for 360 degrees on the exterior surface. The braze need not be flush with the outside surface; however, a single discontinuity must not exceed 1/8″ with a total accumulation of 1/2″ around the periphery. Braze metal must be visible throughout discontinuity when viewed with a 4X magnification and light source.

3.7.1 Joint quality. The brazed joint shall be achieved in a manner which will minimize entrapped slag, voids, unwetted or unbonded areas and other brazing defects. The aggregate area of these defects shall not exceed 15% of the area of the faying surfaces, and the maximum extent of a defect shall not exceed 50% of the overlap distance (width) of the joint.

3.8 Ballistic performance (proving ground). The projectile body assembly shall reveal no evidence of failure when tested at excess pressure. Body assembly failure shall be:

a. Projectile body or fuze adapter deformation attributable to noncompliance with drawing or specification requirements.

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b. Loss or breakup of any component, in the mortar tube or in flight.

c. Short rounds due to undetermined cause.

3.9 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein. (See 4.3 and 6.2).

3.10. Workmanship. All parts and assemblies shall be fabricated in a thorough workmanlike manner. Threads shall be full and undamaged for the entire minimum length as required by the applicable drawings. No part shall have a burr which might be injurious to personnel handling the item. No part or assembly shall contain dirt, grease, chips, rust, corrosion, or other foreign matter. Particular attention shall be given to the cavity of the body assembly, to insure that no grit, scale, accumulated braze alloy, residual flux or other foreign matter remains therein after final cleaning. Projectile assemblies shall be prepared for painting in accordance with TT-C-490. The cleaning method used shall not be injurious to any part nor shall parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48087 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspections set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all

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requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the items as required per Table I. Total quantities are as follows: 10 adapters, 10 projectile bodies (unpainted), 10 cleaned metal parts assemblies less paint, 25 metal parts assemblies after painting, 2 adapter tensile specimens and 4 body tensile specimens.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I First article inspection

MIL-P-64036(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Adapter, dwg. 10534896				
	Examination for defects	10	100%	3.2	4.4.2.1
	Tensile test	2	100%	3.5	4.4.5.3.2
	Body (less paint), dwg. 10534895				
	Examination for defects	10	100%	3.2	4.4.2.2
	Tensile test	1/4	100%	3.5	4.4.5.3.1
	Metal Parts Assembly (prior to paint) dwg. 10534894				
	Examination for defects	10	100%	3.2	4.4.2.3
	Air pressure test	2/10	100%	3.2	4.4.3.2/4.4.5.4
	Braze joint examination	5	100%	3.7.1	4.4.5.5
	Metal Parts Assembly (after painting) dwg. 10534894				
	Examination for defects	25	100%	3.2	4.4.2.4
	Salt spray test	5	100%	3.2	4.4.3.4/4.4.5.6
	Excess pressure test (proving ground)	20	100%	3.8	4.4.3.5/4.4.5.7

NOTES: 1/ Specimens taken from 2 projectiles, 4 specimens total. Specimens may be pre-machined.
2/ After passing air leak test, 5 shall be sectioned for braze joint examination.

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078 and the following:

4.4.1.1 Forging lots. Forging lots shall contain body forgings from not more than one heat of steel, that have been heat treated in the same heat-treatment equipment, at the same temperature, in one unchanged process. The forging lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.2 Cold shaping lots. Cold shaping lots shall contain cold shaped bodies from not more than one heat of steel, that have been cold worked by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. The cold shaping lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.3 Miscellaneous lot. Small lots, such as bodies from abnormally small heats of steel or bodies which have become separated from previously delivered lots, etc., may, at the discretion of the manufacturer, be grouped into miscellaneous, forging, cold shaping or casting lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and the data card so noted. Tension test quantities shall be doubled.

4.4.1.4 Grouping. Forging, or cold shaping production lots shall remain grouped together for lot identity purposes according to forging, casting or cold shaping lots, until such time as the results of the tension test have indicated that the lots have complied with the mechanical properties requirements of the applicable drawing and are suitable for delivery as projectile lots.

4.4.1.5 Projectile lot. All forgings or cold shaping production lots that have been determined to have complied with the mechanical properties requirements of the applicable component drawing, may be re-grouped for assembly into projectile lots. A projectile lot shall consist of projectile assemblies produced by one manufacturer, under one contract, in one unchanged process, in accordance with same drawing, same drawing revision, same specification and same specification revision.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Adapter	1 OF 2	10534896
			NEXT HIGHER ASSEMBLY: 10534894

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective		100%	3.6.2	Visual
2	Diameter of press fit surface improper		100%	3.2	Gage
3	Presence of a reverse taper on press fit surface		100%	3.2	Gage
4	Profile of blend of radius with press fit surface improper		100%	3.2	Gage
5	Surface finish of press fit surface improper		100%	3.2	Gage
6	Radius at start of press fit area missing or improper for full circumference		100%	3.2	Visual (See 6.1)
Major					
101	Diameter of brazing surface		0.40%	3.2	Gage
102	Large outside diameter		0.40%	3.2	Gage
103	Depth from front face to bottom of fuze cavity		0.40%	3.2	Gage
104	Depth of brazing groove, max. (if applicable)		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Adapter (Cont)	2 OF 2	10534896
			NEXT HIGHER ASSEMBLY: 10534894

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
105	Length from front face to projectile shoulder		0.40%	3.2	Gage
106	Pitch diameter of thread		0.40%	3.2	Gage
107	Minor diameter of thread		0.40%	3.2	Gage
108	Effective thread, min.		0.40%	3.2	Gage
109	Tensile test	2	100%	3.2	Gage/4.4.3.1.4
110	Length of flat, min., forward of brazing groove (if applicable)		0.40%	3.2	Gage
Minor					
201	Forward outside diameter		0.65%	3.2	Gage
202	Diameter of chamfer at front		0.65%	3.2	Gage
203	Rear outside diameter		0.65%	3.2	Gage
204	Length, total		0.65%	3.2	Gage
205	Length of rear angle		0.65%	3.2	Gage
206	Angle rear of brazing surface		0.65%	3.2	Gage
207	Radii or chamfer missing or incorrect		0.65%	3.2	Visual/6.1
208	Surface finish improper (except press fit surface)		0.65	3.2	Visual/6.1
209	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Body		10534895
			NEXT HIGHER ASSEMBLY 10534894

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective		100%	3.6	Visual
Special					
a	Profile of gas-check grooves		0.40%	3.2	Gage
c	Minor diameter of rear thread, max.		100%	3.2	Gage
Major					
101	Runout of forward inside diameter with outside diameter of body		0.40%	3.2	Gage
102	Runout of inside diameter with outside diameter		0.40%	3.2	Gage
103	Forward inside diameter		0.40%	3.2	Gage
104	Large inside diameter		0.40%	3.2	Gage
105	Outside diameter, rear of bourrelet		0.40%	3.2	Gage
106	Diameter of bourrelet		0.40%	3.2	Gage
107	Length of brazing surface, min.		0.40%	3.2	Gage
108	Length to outside basic diameter from rear face		0.40%	3.2	Gage
109	Length from rear face to inside reference diameter		0.40%	3.2	Gage
110	Length from front face to rear of bourrelet		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.2	Body (Cont)		10534895
			NEXT HIGHER ASSEMBLY 10534894

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
111	Depth of bourrelet grooves		0.40%	3.2	Gage
112	Wall thickness		0.40%	3.2	Gage
113	Sharp corners not present at front or rear edges of bourrelet grooves or rear of bourrelet		0.40%	3.2	Visual
114	Pitch diameter of rear thread		0.40%	3.2	Gage
115	Length, total		0.40%	3.2	Gage
116	Tensile test	1/4	100%	3.2	Gage/4.4.3.1.3
117	Angle at front outside diameter		0.40%	3.2	Gage
118	Radius at base of cavity		0.40%	3.2	Gage
119	Radius of blend in cavity, max.		0.40%	3.2	Gage
120	Forward outside diameter (O.D.)		0.40%	3.2	Gage
121	Radius of blend at front of taper on O.D.		0.40%	3.2	Gage
Minor					
201	Rear outside diameter		0.65%	3.2	Gage
202	Diameter at rear chamfer		0.65%	3.2	Gage
203	Diameter at rear undercut		0.65%	3.2	Gage
204	Width of undercut		0.65%	3.2	Gage
205	Depth of thread cavity		0.65%	3.2	Gage
206	Angle at rear face		0.65%	3.2	Gage
207	Angle at rear of braze surface		0.65%	3.2	Gage
208	Radii or chamfers missing or incorrect		0.65%	3.2	Visual/6.1
209	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: 1/ Two projectiles selected, four specimens total.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Projectile, 60MM, SMOKE, WP, M302, Metal Parts Assembly (Prior to Paint)	1 OF 2	10534894
			NEXT HIGHER ASSEMBLY 9215575/12561290

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Air pressure test	5	100%	3.2	Visual
2	Braze joint defective		100%	3.7.1	Visual/4.4.5.5
Special					
a	Braze metal not visible around entire periphery of joint		100%	3.2	Visual
Major					
101	Runout of pitch diameter of fuze thread with outside diameter (O.D.) of body and front face		0.40%	3.2	Gage
102	Runout of bourrelet with body O.D.		0.40%	3.2	Gage
103	Runout of press fit surface with body O.D.		0.40%	3.2	Gage
104	Runout of adapter with body O.D.		0.40%	3.2	Gage
105	Runout of rear taper with body O.D.		0.40%	3.2	Gage
106	Runout of pitch diameter of rear thread with body O.D. and rear face		0.40%	3.2	Gage
107	Presence of flux, oxides or foreign matter in cavity or on exterior		0.40%	3.2	Visual
108	Protective finish missing or inadequate		0.40%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.3	Projectile, 60MM, SMOKE, WP, M302, Metal Parts Assembly (Prior to Paint)				10534894
					NEXT HIGHER ASSEMBLY
					9215575/12561290
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Marking missing, incorrect or illegible		0.65%	3.2	Visual
202	Presence of protective coating on front or rear face		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Projectile, 60MM, SMOKE, WP, M302, Metal Parts Assembly (After Paint)	1 OF 1	10534894
			NEXT HIGHER ASSEMBLY: 9215575/12561290

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective		100%	3.6	Visual
Major					
101	Weight		0.40%	3.2	Gage
102	Diameter of bourrelet max.		0.40%	3.2	Gage
103	Protective coating in cavity inadequate or excessive		0.40%	3.2	Visual
104	Paint or varnish on threads		0.40%	3.2	Visual
105	Excess pressure test (proving ground)	10	100%	3.8	4.4.3.5/4.4.5.7
106	Corrosion preventive compound missing from threads or press fit surface		0.40%	3.2	Visual
107	Total length, max		0.40%	3.2	Gage
108	Salt spray test	5	100%	3.2	Visual/4.4.3.4
Minor					
201	Paint present on front or rear surfaces		0.65%	3.2	Visual
202	Paint pooled or wet		0.65%	3.2	Visual
203	Exterior paint inadequate or damaged		0.65%	3.2	Visual
204	Corrosion inhibitor (crystal) or VTI tablet missing from body cavity		0.65%	3.2	Visual
205	Weight of corrosion inhibitor incorrect		0.65%	3.2	Gage
206	Protective cap missing or damaged		0.65%	3.2	Visual
207	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Box, Fiberboard, Ammunition for Interplant Shipment of Projectile, Empty (Before Sealing)	1 OF 1	7548319
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Support missing or damaged		0.65%	3.2	Visual
202	Separator missing or damaged		0.65%	3.2	Visual
203	Improper assembly		0.65%	3.2	Visual
204	Incorrect number in carton		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Box, Fiberboard, Ammunition for Interplant Shipment of Projectile, Empty (After Sealing)	1 OF 1	7548319
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box severely damaged, cut or punctured through all layers		0.40%	3.2	Visual
Minor					
201	Box damaged, cut or punctured		0.65%	3.2	Visual
202	Box not properly sealed		0.65%	3.2	Visual
203	Contents loose		0.65%	3.2	Manual
204	Marking missing, incorrect or illegible		0.65%	3.2	Visual

NOTES:

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4.4.3 Testing. The following tests shall be performed in accordance with the provisions of Table I and 4.4.

4.4.3.1 Mechanical properties.

4.4.3.1.1 Hardness (forgings). See 4.4.1.1. In order to determine the body forgings that will be used for the tension test, the base of each body forging in the forging lot, after heat treatment and prior to machining, shall be subjected to a hardness test with a Rockwell Hardness Tester or by any other method of hardness testing approved by the procuring activity.

4.4.3.1.2 Hardness (cold shapings). See 4.4.1.2 in order to determine the bodies formed by cold shaping that will be used for the tension test, the base of 25 bodies in the cold shaping lot, after stress relief and prior to machining, shall be subjected to a hardness test with a Rockwell Hardness Tester or by any other method of hardness testing approved by the procuring activity. See 4.4.5.3.1.

4.4.3.1.3 Tensile test. From 4.4.3.1.1 and 4.4.3.1.2, two bodies, one of the softest and one of the hardest from each forging or cold shaping lot (see 4.4.1) shall be subjected to this test to determine compliance with the mechanical properties required by the applicable drawing. Failure of one or more tension test specimens to comply with the requirements shall be cause for rejection of the lot representative of the specimen. See 4.4.5.3.1.

4.4.3.1.4 Adapter. One tension-test specimen shall be taken from each of 2 bars representative of the production lot to determine compliance with yield strength and elongation requirements of the applicable drawing. Failure of either specimen to comply shall be cause for rejection of the lot. See 4.4.5.3.2.

4.4.3.2 Air pressure test. Each body assembly shall be subjected to a submerged air pressure test to determine compliance with the assembly drawing. Failure of any assembly to comply as evidenced by the presence of air bubbles in the vicinity of the brazed joint or rising to the surface of the water shall be cause for rejection of the assembly. Body assemblies which comply shall immediately upon removal from the test fixture be stamped with the letter “A” as indicated on the assembly drawing. See 4.4.5.4.

4.4.3.3 Joint quality. After air pressure testing, but prior to the application of protective coating, body assemblies shall be

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selected at random and sectioned to determine compliance with 3.7.1. Failure of any specimen to comply shall be cause for rejection of the lot. See 4.4.5.5.

4.4.3.4 Salt spray test. The specimens submitted for testing in accordance with Table I and 4.4.2.4 shall be tested in accordance with the requirements of dwg. 10534894 and 4.4.5.6 herein.

4.4.3.5 Proving ground. Twenty sample projectiles shall be selected for this test. The test shall be performed as specified in 4.4.5.7. Defect classification and individual accept/reject criteria shall be in accordance with the following table:

TABLE II Proving ground test

Categories	Defect	Accept/Reject	
Critical:	Premature burst, metal part break-up, or deformation due to launch	0	1
Major	Horizontal range less than 80% of average range	1	2

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.4.5 Methods of inspection.

4.4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of the specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing, specification or specification requirements, and that all specified manufacturing processes have been followed.

4.4.5.2 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Section 3, 4, and 5 of the specification in accordance with 4.4.2 herein.

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4.4.5.3 Tension test.

4.4.5.3.1 Projectile body. From bodies selected in accordance with Table I and 4.4.3.1.3, two tension test specimens, conforming to the "T" type specimen prescribed in figure 1 of appendix 1, shall be taken longitudinally from each side of the body 180 degrees apart. Equipment and procedures shall be as prescribed by ASTM E-8.

4.4.5.3.2. Adapter. Tension test specimens shall conform to the largest standard round specimen obtainable in accordance with ASTM E-8. The test shall be performed using equipment and procedures prescribed by ASTM-E8. See 4.4.3.1.4.

4.4.5.4 Air pressure test. Each projectile shall be cleaned of dirt, grease, oil, chips or other foreign matter. Immediately after the cleaning operation each projectile shall be assured to be dry, and then the air pressure test will be conducted. This test shall be made using equipment capable of performing the test accurately and consistently. Body assemblies shall be mounted in an approved fixture in such a manner that the braze joint shall be submerged in water, then the required air pressure shall be applied and held for a minimum of 15 seconds. The water in the tank shall be permitted to settle after immersion of the assemblies prior to the start of timing. Observation for leakage shall be made for 360 degrees of periphery for the entire time the assembly is required to be pressurized. The equipment shall be calibrated at the start of each day's operation, at intervals of 4 hours and after a stoppage of 3 hours or more.

4.4.5.5 Joint quality. The samples, selected in accordance with Table I and 4.4.3.3., shall be sectionalized as follows:

(a) The body portion of the projectile assembly to the rear of the nose adapter braze shall be removed in such a manner that no portion of the braze joint will be damaged.

(b) The remaining portion of the body containing the nose adapter shall then be sectioned into four segments by cutting vertically through the adapter and body.

(c) The brazed areas of the four pieces of brazed adapter and body shall be visually examined to determine compliance with 3.7.1. Failure of any sample shall cause referral of the lot.

4.4.5.6 Salt spray test. The test shall be conducted using component parts painted at the same time in the same equipment as other deliverable parts. The salt solution, equipment, preparation of test parts, test procedures and drying time shall be in accordance with MIL-STD-810 Method 509.2 (salt fog). After drying, any evidence of paint failure, non-adhesion or corrosion on a

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surface painted with protective finish shall cause rejection of all parts painted during that operation. See 4.4.3.4.

4.4.5.7 Proving ground test.

4.4.5.7.1 Excess pressure. The samples supplied in accordance with Table I and 4.4.2.4 (inert loaded to the service weight and dummy fuzed) shall be assembled into complete rounds and fired for recovery in a mortar for which the round is standard, the tube of which will have a minimum of 25 percent life remaining based on current mortar tube condemnation criteria and propellant charge adjusted to produce 6000 psi minus 350 psi. Photographic equipment shall be set up near the muzzle to record the flight of the projectile to help determine compliance. The test shall be performed at a Government Proving Ground in accordance with applicable USA TECOM acceptance test procedures. Cartridges shall be conditioned at 70 degrees plus or minus 5 degrees Fahrenheit for 4 hours prior to firing.

5. PACKAGING

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with the requirements of Level C in MIL-STD-1169 and dwg. 7548319.

6. NOTES

6.1 Visual examination qualification. When a compliance with the applicable requirement is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of test data. The following addresses are to be added to the Ballistic Test Request (BTR) for distribution of firing test reports to the technical activity: Commander, US Army Armament, Research, Development and Engineering Center (ARDEC), ATTN: AMSMC-QAT-B(D) and SMCAR-CCH-P, Dover, NJ 07801-5001.

6.4 Submission of inspection equipment designs for approval. See para 6.2.3 of MIL-A-48078. Submit designs as required to: Commander, US Army Armament, Research, Development and Engineering Center (ARDEC), ATTN: AMSMC-QAT-I, Dover, NJ 07801-5001.

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6.5 Drawings. Drawings listed in Section 2 of this specification under the headings US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or US Army Armament Research and Development Command (ARRADCOM) drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

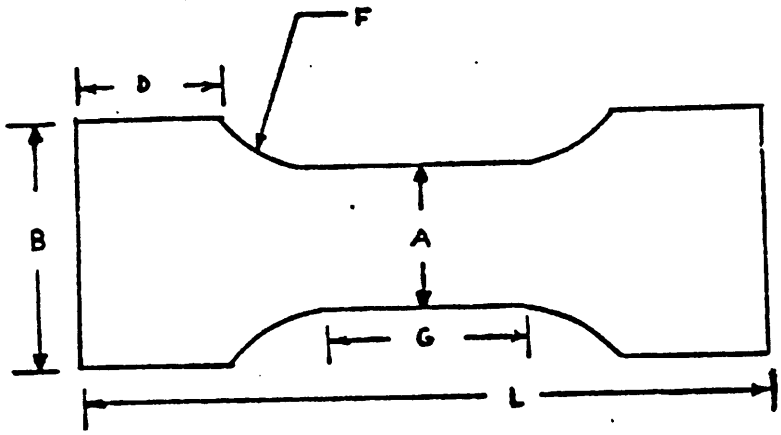
6.6 Supersession data. This specification includes the requirements of Frankford Arsenal Procurement and production Directorate FA-PD-MI-2640 Rev. O dated 15 February 1965, and Amendment 5 dated 13 March 1973.

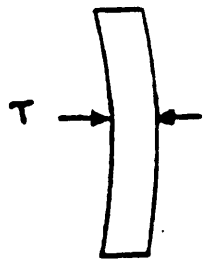
6.7 Subject term (key word) listing.

Ammunition
Bourrelet
Brazing
Casting
Faying
Forgings
Mortars
Projectiles
Shaping, cold
Slag
Steel, heat of

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LONGITUDINAL SPECIMEN





CROSS SECTION

A - Width At center	- 3/4 ± .010	inch
B - Width At Grip End	- 1 1/4	"
D - Grip Length (Approx.)	- 3/4	"
F - Radius (Min.)	- 1/2	"
G - Gage Length	- 1 ± .005	"
L - Total Length (Approx.)	- 3 1/2	"

FIGURE 1

TENSILE TEST SPECIMEN

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Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1310-A383)

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-64036 (AR)	PROJECTILE, 60MM, SMOKE, WP, M302 METAL PARTS ASSEMBLY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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